

Nicholas A. Ball Ulmer

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HEADLINE

Software engineering student focusing on software architecture with experience in financial data science. Proficient in Java for backend services and databases with a Google certificate in data analysis.

EDUCATION

Georgia Institute of Technology, Atlanta, GA Aug 2025 – May 2027
Master of Science in Computer Science: Artificial Intelligence Track GPA: 4.0/4.0

Relevant Coursework:

Machine Learning for Trading - Market simulator, trade signal modeling, strategy backtesting
Software Architecture & Design - Modular architecture, design patterns, systems design

Northwest Missouri State University, Maryville, Missouri Aug 2019 – Dec 2022
Bachelor of Science in Mathematics GPA: 3.7/4.0
Purdy Emerging Leaders Scholarship Recipient

PROJECTS

OvO: Power User Oriented Cycle Tracking App. *Kotlin, Android, SQLite* March 2026

- Cycle recording & prediction via mobile SQL database and machine learning to optimize prediction error and deviation
- Data-pipeline with rolling window, outlier detection, and recency weighted mean deviation to predict cycles
- Achieved a MAD-normalized prediction error of 1.08, indicating error comparable to data variability

Rhythm Kana: Mobile Rhythm Game. *C# in Unity, JSON Serialization, ScriptableObjects* January 2026

- From scratch language learning service using cleaned language data from phoneme segmentation & lexical extraction
- Applied dependency inversion to decouple input from game state and game state from core logic and data.
- Adhered to SOLID principles for minimum coupling, maximum modularity and single responsibility

Kinetic Arena: Video Game Design Project. *C# in Unity, Git* Fall 2025

- Worked with a team to produce a small game with 3D environment, enemies, and game feel
- Implemented navmesh navigation and finite state machine logic for enemy AI
- Used Git version control to combine AI systems with team's enemies, health, damage, collisions, and game logic systems

Trip Data Extraction & Analytics Pipeline. *R: Studio, Tidyverse, Google Maps Distance Matrix API* Fall 2024

- Created an automated trip and mileage logging system for business mileage tracking and tax reporting
- R packages to extract image text, clean data, parse addresses, and use Google's Distance Matrix API to compute mileage
- Resulting project allows single click and input of image data to instantly create an IRS compliant mileage log

EXPERIENCE

Data Science Research Assistant Jan 2021 – Dec 2022

Northwest Missouri State University, School of Mathematics and Statistics

- Explored growth vs. fixed mindset data of university students to evaluate the impact of teaching methods on mindset
- Performed large dataset cleaning, aggregation, and exploratory analysis to compare groups and do distribution analysis
- Made insights to the data and presented research at Southwest Educational Research Association conference Feb. 2022

East Asia Organization: Cofounder & President Jan 2020 – Dec 2022

Northwest Missouri State University

SKILLS

Programming: Java, Python, Kotlin, C#, R, SQL
Tools & Platforms: Git, Docker, React, BigQuery, Unity, Linux
Tech: Full Stack, Agile